

PAGE_14_RESEARCH_REPORTS

URL: <https://securityblogs.com.au/knowledge-hub/research-reports/> **Airtable table:** SB · 14 · Research Reports **Page kind:** listing **Sections in spec:** 4

Business Purpose

Long-form research deliverables, gated-download capable.

Sections (top to bottom)

#	Section	Block type
1	Hero	hero
2	Report cards	grid (server)
3	Download CTAs	rich-text
4	CTA band	cta-band

Airtable Tables

Table	Use
SB · 14 · Research Reports	Per-section spec for /knowledge-hub/research-reports/ (one row per section)
SB · 00 · Pages Index	Master index entry
SB · DOCS · Index (proposed)	Links this MD/PDF to the Airtable page record

PostgreSQL Tables (used by this page)

Table	Use
posts	Listing + post cards
media	Image attachments (hero, covers, logos)

API Endpoints

Endpoint	Method	Cached	Purpose
GET /api/posts?where [category][equals]=research-reports	GET	ISR 60s	Post list
GET /api/globals/settings	GET	ISR 600s	Brand + contact + footer

Endpoint	Method	Cached	Purpose
GET /api/redirects?where[isActive][equals]=true	GET (middleware)	60s mem	Redirect rules
POST /api/leads	POST	dynamic	Form submission

Request Schemas

POST /api/leads

```
{
  "name": "string (1-120 chars, required)",
  "email": "string (valid format, required)",
  "phone": "string (optional)",
  "company": "string (optional)",
  "subject": "string (optional)",
  "message": "string (optional, max 2000 chars)",
  "source": "research-download",
  "meta": { "reportSlug": "..." },
  "cf-turnstile-response": "string (Cloudflare token, required in production)",
  "company_url": "string (honeypot – must be empty)"
}
```

Response Schemas

GET /api/pages?where[slug][equals]=...

```
{
  "docs": [{
    "id": "rec...",
    "slug": "research-reports",
    "title": "Research Reports",
    "modules": [
      { "blockType": "hero", "h1": "...", "subtitle": "...", "ctas": [...] },
      { "blockType": "rich-text", "title": "...", "body": { /* Lexical JSON */ } }
    ],
    "status": "published"
  }],
  "totalDocs": 1
}
```

POST /api/leads

```
// 200 OK
{ "ok": true, "id": "rec..." }

// 400 Bad Request
{ "ok": false, "error": "Email looks invalid" }

// 429 Too Many Requests
{ "ok": false, "error": "Too many submissions. Please wait 15 minutes and try again." }

// 502 Bad Gateway (CMS write rejected)
{ "ok": false, "error": "Could not save your enquiry. Please try again." }
```

React Components

Component	Purpose
app/knowledge-hub/research-reports/page.tsx	Server component — fetches CMS data, renders
components/layout/Navbar.tsx	Component used on this page
components/layout/Footer.tsx	Component used on this page
components/layout/Navbar.tsx	Shared header
components/layout/Footer.tsx	Shared footer

Node.js Services

Service / module	Purpose
lib/cms.ts	Typed Payload REST client — all reads
middleware.ts	Edge middleware — CMS redirect rules
lib/submitForm.ts	Browser-side POST helper
app/api/leads/route.ts	Server endpoint — Turnstile + rate-limit + Lead create
cms/src/collections/Leads.ts	Lead schema + afterChange email hook
cms/src/email/transport.ts	Hostinger SMTP via nodemailer

Environment Variables

Var	Required?	Purpose
NEXT_PUBLIC_SITE_URL	Yes	Canonical site URL
CMS_URL	Yes	Where lib/cms.ts fetches from
PAYLOAD_API_KEY	Yes (prod)	Server-side CMS write auth
NEXT_PUBLIC_TURNSTILE_SITE_KEY	Yes (prod)	Cloudflare widget site key
TURNSTILE_SECRET_KEY	Yes (prod)	Server-side verify
SENTRY_DSN	Yes (prod)	Error tracking

Validation Rules

/api/leads validation pipeline

1. **Honeypot:** company_url must be empty. If filled, return { ok: true } silently (do NOT create lead).
2. **Shape:** name and email required. email must match /^[^@\\s]+@[^@\\s]+\\. [^@\\s]+\$/.
3. **Rate limit:** Per-IP, 15 submissions per 15 minutes. Returns 429 over limit.
4. **Turnstile:** Server-side siteverify against Cloudflare. Skipped if TURNSTILE_SECRET_KEY unset (dev mode).
5. **Source:** Must be one of the SOURCE_OPTIONS values defined in cms/src/collections/Leads.ts.
6. **Forensic:** IP, user-agent, referer captured automatically.

Source-specific validation runs in the Next.js layer before the CMS write (zod schemas).

Authentication Rules

- Public POST allowed for browser-side form submission.
- Bot protection: honeypot + Turnstile + per-IP rate limit (15/15min).
- Server-to-server CMS write uses `Authorization: users API-Key <PAYLOAD_API_KEY>`.
- No NextAuth session required from the visitor.

Deployment Notes

- Server-rendered Next.js page (ISR). NOT compatible with `output: 'export'`.
- First deploy requires `pnpm payload migrate + relevant pnpm seed:*` before booting.
- Caddy reverse-proxies port 3000 (marketing) — see `DEPLOYMENT_GUIDE` §6.
- Cloudflare Turnstile site key + secret must be configured in production env.
- Hostinger SMTP credentials must be set in `cms/.env` for assignee notification emails.

Testing Procedures

Local smoke

1. `pnpm dev` (marketing site at `http://localhost:3000`)
2. Navigate to `/knowledge-hub/research-reports/`
3. Verify hero renders, navigation works, no console errors
4. Fill the form with valid data → expect 200 OK + lead in `/admin/collections/leads`
5. Fill the form with empty email → expect 400
6. Submit 16 times in 15 min from same IP → 16th returns 429
7. Open with JS disabled → confirm graceful no-JS fallback message

Production checks

- Lighthouse Performance ≥ 85 on landing pages
- Lighthouse Accessibility ≥ 95
- Lighthouse SEO = 100
- Sentry: no new error fingerprints after deploy
- Uptime monitor green

Troubleshooting

Symptom	Likely cause	Fix
Page returns 500	CMS unreachable	Check <code>CMS_URL</code> + CMS service status
Content stale after admin edit	Cache not purged	<code>revalidateTag()</code> or wait for revalidate interval
Sitemap missing this page	CMS Pages record not published	Set <code>status=published</code>

Symptom	Likely cause	Fix
Navbar/footer shows defaults	Settings global missing	Run <code>pnpm seed:settings</code>
Form submits but no Lead	PAYLOAD_API_KEY wrong	Regenerate in admin → Users → API Key
Form rejected with 400 captcha	Turnstile token bad	Verify production TURNSTILE_SECRET_KEY matches site key
Assignee not getting email	SMTP creds wrong	Test Hostinger SMTP from VPS shell

Change Impact Analysis

If you change...

- **Copy / hero text:** Low impact. Edit the CMS record (or `app/knowledge-hub/research-reports/page.tsx` if still hardcoded). Wait for ISR.
- **Sections order:** Medium impact. Re-order modules in admin. Renders instantly on next ISR.
- **Add a new module type:** High impact. Add block to `cms/src/collections/Pages.ts`, add component to `components/modules/blocks/`, add dispatch in `ModuleRenderer.tsx`.
- **Slug: CRITICAL — breaks every URL.** Add a `Redirects` record from old slug → new BEFORE updating. Notify SEO if any external links exist.
- **Authentication scope:** Edit `cms/src/access/*.ts`. Roles changes only by `super_admin`.
- **API endpoint shape:** Breaking change. Bump version, dual-route for 30 days, document migration.

Rollback Procedures

1. **CMS content rollback:** Use Payload versions UI (`/admin/collections/pages/<id>` → Versions tab). Revert to a prior snapshot.
2. **Code rollback:** `git revert <bad-commit>` on `phase-c-frontend-rewire`, push, redeploy.
3. **Schema migration rollback:** Restore Postgres from nightly backup (see `DEPLOYMENT_GUIDE` §7).
4. **Deploy rollback:** `systemctl stop sb-web && git checkout <prev-commit> && pnpm build && systemctl start sb-web`. <5 min.
5. **DNS-level rollback (emergency only):** Revert A record to the last static-export Hostinger IP (5-minute propagation).

Forms

- **Download form (email gate)** — posts to `/api/leads` (see Request/Response schemas)

Notes: Gated-download form posts to `/api/leads` with `source='research-download'`.

End of PAGE_14_RESEARCH_REPORTS.md